

b) Vulnerability scanning vs Penetration Testing

Vulnerability Scanning	Penetration Testing
Identifies known vulnerabilities in a system.	Attempts to exploit vulnerabilities to assess security risks.
Usually automated using scanning tools.	Often involves manual testing and exploitation.
Provides a list of weaknesses.	Demonstrates the actual impact of weaknesses.
tool: WPScan	Tool: Metasploit.

Section B: Practical Task

Task: Network Scanning & Analysis

Step 1: Host Discovery

Nmap -sn 172.16.173.0/24

Nmap -sn tells nmap to discover which hosts are online without performing a port scan.

```
Session Actions Edit View Help
(blueteam@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether 00:0c:29:c2:7e:cd txqueuelen 1000 (Ethernet)
    RX packets 9 bytes 846 (846.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 47 memory 0x3fe00000-3fe20000

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether 00:0c:29:c2:7e:e1 txqueuelen 1000 (Ethernet)
    RX packets 9 bytes 846 (846.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 47 memory 0x3fd00000-3fd20000

eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.16.173.138 netmask 255.255.255.0 broadcast 172.16.173.255
    inet6 fe80::20c:29ff:fec2:7ed7 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:c2:7e:d7 txqueuelen 1000 (Ethernet)
    RX packets 8 bytes 960 (960.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 32 bytes 3634 (3.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 56 memory 0x3e600000-3e620000

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 4 bytes 200 (200.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 4 bytes 200 (200.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(blueteam@kali)-[~]
$ sudo nmap -sn 172.16.173.0/24
[sudo] password for blueteam:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-12 10:46 +0300
Nmap scan report for 172.16.173.1
Host is up (0.00020s latency).
MAC Address: 3E:06:30:33:AE:65 (Unknown)
Nmap scan report for 172.16.173.2
Host is up (0.00053s latency).
MAC Address: 00:50:56:E7:62:92 (VMware)
Nmap scan report for 172.16.173.254
Host is up (0.00051s latency).
MAC Address: 00:50:56:EC:32:74 (VMware)
Nmap scan report for 172.16.173.138
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.04 seconds

(blueteam@kali)-[~]
```

Step 2: Port Scanning (2 Marks)

Scan for open ports and services:

`nmap -sS -sV 172.16.173.138` -sS-TCP SYN port scan -sV-service/version detection

```
(blueteam@kali)-[~]
└─$ nmap -sS -sV 172.16.173.138
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-12 11:58 +0300
Nmap scan report for 172.16.173.138
Host is up (0.00036s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 10.3p1 Debian 4 (protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.00 seconds

(blueteam@kali)-[~]
└─$
```

Step 3: OS Detection

`Nmap -A 172.16.173.138` – Aggressive scan

```
(blueteam@kali)-[~]
└─$ sudo nmap -O 172.16.173.138
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-12 12:01 +0300
Nmap scan report for 172.16.173.138
Host is up (0.00015s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Linux 5.0 - 6.2 (96%), HP P2000 G3 NAS device (92%), Linux 3.7 - 4.19 (91%), Linux 2.6.32 (91%), Linux 4.1 (91%), Linux 4.10 (91%), Android 7.0 (Linux 3.18) (90%), Linux 3.8 - 3.9 (90%), Linux 5.4 - 5.8 (90%), Linux 6.17 (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.25 seconds

(blueteam@kali)-[~]
└─$ sudo nmap -A 172.16.173.138
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-12 12:02 +0300
Nmap scan report for 172.16.173.138
Host is up (0.00011s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 10.3p1 Debian 4 (protocol 2.0)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Linux 5.0 - 6.2 (96%), HP P2000 G3 NAS device (92%), Linux 3.7 - 4.19 (91%), Android 7.0 (Linux 3.18) (91%), Linux 2.6.32 (91%), Linux 4.1 (91%), Linux 4.10 (91%), Linux 3.8 - 3.9 (91%), Linux 5.4 - 5.8 (91%), Linux 6.17 (91%), Linux 4.10 (91%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 0 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

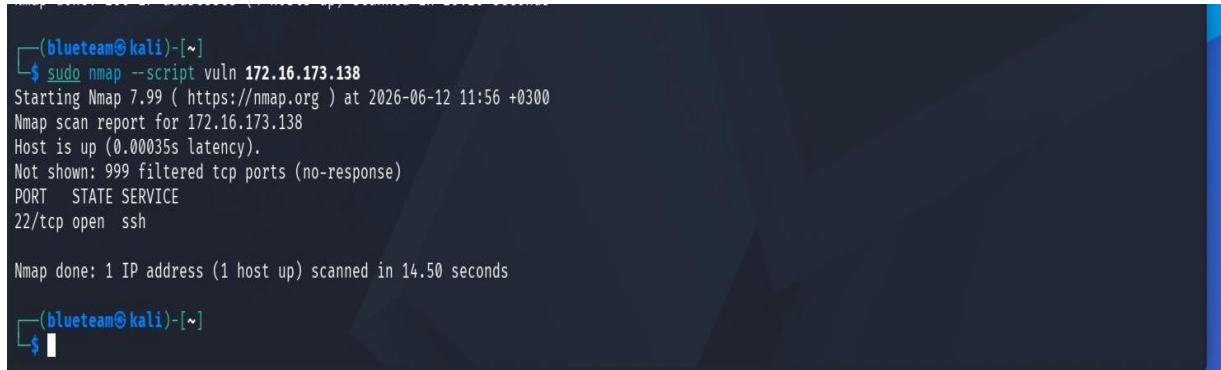
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.06 seconds

(blueteam@kali)-[~]
└─$
```

Step 4: Basic Vulnerability Scan

Run a basic vulnerability detection script:

```
nmap --script vuln 172.16.173.138
```



```
(blueteam@kali)-[~]  
$ sudo nmap --script vuln 172.16.173.138  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-12 11:56 +0300  
Nmap scan report for 172.16.173.138  
Host is up (0.00035s latency).  
Not shown: 999 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
  
Nmap done: 1 IP address (1 host up) scanned in 14.50 seconds  
  
(blueteam@kali)-[~]  
$
```

Section C: Analysis & Reporting

Network Scanning and Analysis Report

A network scan was performed on the localhost address **172.16.173.138** using Nmap in Kali Linux. The host discovery scan confirmed that the system was active and reachable. During the port scanning phase, open ports were identified. Common port discovered included SSH (port 22) indicating that remote access is running on the system.

The service version scan revealed the specific software versions associated with the open port. This information is useful for identifying outdated services that may contain known security weaknesses. The operating system detection feature indicated that the target machine was running **(91%) Linux-based operating system**.

A vulnerability scan was then conducted using the Nmap vulnerability scripts. The scan checked the detected services against known vulnerabilities. The target host is reachable and has one exposed service (SSH on port 22). All other TCP ports are filtered, indicating that a firewall or network filter is blocking access. No additional services were detected during the scan.

Nmap scripts due to the limited number of exposed services identified no direct vulnerabilities.

Security Assessment

Risks Identified SSH exposed (port 22) Remote access enabled, Potential brute-force target (if weak credentials exist) limited visibility, Firewall filtering most ports could hide additional services.

Medium Observations.

Service version visible (OpenSSH 10.3p1) OS fingerprinting partially exposed

Positive Security Indicators

No unnecessary services running Strong port filtering (attack surface reduced)

No vulnerabilities detected via Nmap scripts

Recommendations

SSH Hardening-Disable password authentication, Use SSH key-based login only Restrict SSH access to trusted IPs Enable intrusion prevention.

Network Security-Keep firewall filtering active (good practice)Monitor SSH logs for brute-force attempts

Monitoring Improvements-Enable logging for connection attempts